Diego Alducin, Ph.D.

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SUMMARY

Senior Data Scientist with a Ph.D. and a multidisciplinary background in **physics, computer science, and software engineering**, specializing in the development of **scalable machine learning models and predictive analytics**. Proven track record leading AI/ML initiatives in **regulated, high-impact domains**, including **cybersecurity, autonomous systems, and space technologies**. Skilled in **statistical modeling, time-series forecasting, object detection**, and deploying solutions aligned with risk and compliance standards. Extensive experience with **Python, SQL, AWS, and data pipeline engineering**, building models that power automation, risk mitigation, and business intelligence. Adept at collaborating across teams to translate complex challenges into **production-ready, explainable models**, and mentoring junior engineers and scientists to elevate team capability. Passionate about applying advanced analytics to enable **financial resilience, operational optimization**, and long-term business value.

EXPERIENCE

Senior Computer Scientist

Southwest Research Institute **October 2024 - Present, San Antonio, TX**

- Leading software development for a high-reliability satellite subsystem using **C++**, ensuring robust performance in mission-critical environments.
- Developed an algorithm using **object detection**, **random forests** and **advanced data analysis** to identify specific events from the Juno Mission with 94% accuracy.
- Managing a $350k project with 3 engineers to investigate the effects of Electromagnetic Interference (EMI) on electric vehicles for the National Highway Traffic Safety Administration (NHTSA).
- Continually developing business relationships with customers to provide cutting-edge solutions, and leading efforts to obtain funding for new projects and technologies.
- Leading a team of 4 engineers to develop **Large Language Model (LLM)** to improve the efficiency of cybersecurity testing.

Research Computer Scientist

Southwest Research Institute **January 2023 - October 2024, San Antonio, TX**

- Implemented a secure micropatching algorithm for communications between space systems successfully deployed in the International Space Station (**ISS**) on an Amazon Web Services (**AWS**) Snowcone.
- Contributed to developing cutting-edge technology in the **DARPA** CRASH program.
- Developed a web application that uses **React** and **Redux** to create cybersecurity test plans for automotive systems.
- Led the creation of an **API** using **Python** to reverse engineer CAN bus messages from vehicles directly from the OBDII port with 91% accuracy using clustering and regression analysis.
- Created a **Long short-term memory (LSTM) Neural Network** that predicts the path and velocity of a vehicle based on Basic Safety Messages with 89% accuracy.

Expert Failure Analysis Engineer

Tower Semiconductor **May 2019 - January 2023, San Antonio, TX**

- Established the area of Failure Analysis within the fab to streamline requests between departments, improving analysis turnaround time by 80%.
- Created a ticketing database web system with **SQL**, **JavaScript**, **PHP** and **Python** that processed requests, set priorities, and collected data from analysis.
- Conducted in-depth failure analysis of diverse microelectronic device families in high-volume manufacturing environments using advanced diagnostic techniques and predictive analytics such as **regression analysis** and **support vector machines**.
- Correlated failure mechanisms of technologies between electrical measurements, process development, and physical analysis using **Python** and KLA Data Analysis Environment.
- Involved in quality audits for customers to determine specific production process requirements.
- Negotiated and obtained capital, $600k, for state-of-the-art instruments and upgrades.
- Deployed cutting-edge failure analysis workflows, integrating advanced diagnostic techniques to boost issue resolution efficacy by 30% in the semiconductor sector.

Senior Process Engineer

Tower Semiconductor **December 2016 - May 2019, San Antonio, TX**

- Supported processes of 12 different technologies for large-scale production of 8-inch wafer semiconductors.
- Identified a cost-saving measure using **classification algorithms** which led to the introduction of a new tool to improve the lifetime of testing materials saving $750k within a year.
- Utilized high-level data analysis using **regression** and **clustering** algorithms to determine the contribution of process changes for chemical, physical, and environmental factors in products for yield improvement and technology transfers.
- Spearheaded a team in integrating data analytics platforms using **Pandas** and **SQL**, leading to a $1.4 Million reduction in test wafer costs.
- Developed new processes for technology transfers with SIPOC documentation.
- Improved process control above a Cpk of 1.67 of critical tools by introducing continuous VPD ICP-MS measurements to accurately detect impurities in chemical concentrations..

EDUCATION

M.Sc. Computer Science

University of Texas at San Antonio • 2024

- Coursework: Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Software Engineering, Cloud Computing
- Final Project: "Physics-Informed Learning for Scalable Single-Qubit Control Simulations".

Ph.D. Physics

University of Texas at San Antonio • 2016

- Coursework: Holography, Materials Science, Nanotechnology, Semiconductor Physics, Instrumentation (Lab-View), Diffraction Physics, Mathematical and Computational Physics.
- Thesis Topic: "In situ Transmission Electron Microscopy Experimentation of Nanostructured Materials.".

B.Sc. Physics

University of Texas at San Antonio • 2012

SKILLS

Web Development & Cloud Technologies: HTML5, CSS, React, Flask, Docker, AWS, GCP.

Programming & Scripting: Python, C, C++, Java, JavaScript, SQL, Bash

Machine Learning & Data Science: Pandas, Numpy, PyTorch, TensorFlow, Keras, Scikit-learn, NLP, Computer Vision, Physics-Informed Neural Networks (PINNs), Object Detection, Anomaly Detection, Ollama, Huggingface, LLMs.

Data Engineering & Big Data: Apache Spark, Apache Airflow, Pandas, SQL.

Technical Documentation: Markdown, LaTeX, Microsoft Office (Word, Excel, PowerPoint, Access).
Languages: English (Native), Spanish (Native).